⚡ SSC CGL — Arithmetic Trick Decision Matrix

Recognition → Shortcut → Formula → Fastest route. Built from the repository's PYQ pattern banks (2021–2025).
Target post: ASO · Tier-I + Paper-I scope · Solve each type in under 30 seconds.

🟢 Fastest method 🔴 Common trap / mistake 🔷 Butterfly (cross-multiply) trick 🔴 High-repeat topic

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How to read every row: Match the question to a **Situation** → confirm via the **Recognition** cue (the words that give it away) → apply the **Shortcut/Formula** → the green column is the single fastest route in the exam → dodge the red trap. The Butterfly and **Ratio** columns are the two speed-tools that recur across almost every topic.

1 · Profit & Loss 🔴 High Repeat · 2–4 Q/paper

Golden rule: fix **CP = 100** (or CP = ₹100 per unit) and everything becomes a percentage you can read off directly.

SITUATION	GIVEN	ASKED	RECOGNITION CUE	SHORTCUT / FORMULA	RATIO / BUTTERFLY TRICK	WORKED EXAMPLE (PYQ STYLE)	FASTEST ROUTE	COMMON MISTAKE
Basic P%/L%	CP & SP	Profit or loss %	"bought … sold …"	`P% = (SP−CP)/CP ×100`	Set CP=100; SP−100 = P%	CP 500, SP 600 → 100/500 = **20%**	**Read gain over CP=100**	Dividing by SP instead of CP
Find CP from SP	SP & P% (or L%)	CP	"gains x%, find CP"	`CP = SP×100/(100+P%)`	SP×(100/(100±%)) as a clean fraction	SP 78, +30% → 78×10/13 = **₹60**	**Turn 100/130 into 10/13, multiply**	Subtracting 30% of 78

SITUATION	GIVEN	ASKED	RECOGNITION CUE	SHORTCUT / FORMULA	RATIO / BUTTERFLY TRICK	WORKED EXAMPLE (PYQ STYLE)	FASTEST ROUTE	COMMON MISTAKE
Mark-up + discount	x% above CP, y% discount	Net profit %	"marks … above, gives … discount"	`x − y − xy/100`	MP=140, SP=140×0.75 on CP 100	+40% mark, 25% disc → 140×.75=105 → **5%**	**Successive-% formula in one line**	Just doing 40−25 = 15%
Same SP, ±x%	2 items, +x% & −x%	Overall %	"one at gain x%, other at loss x%"	`Always LOSS = x²/100 %`	—	±10% → 100/100 = **1% loss**	**Answer is loss $x^2/100$ by memory**	Thinking it nets to 0%
Loss = SP of n articles	"loses SP of n on selling N"	Loss %	"loses = SP of 3 articles"	Let SP=1; CP(N)=N+n; `Loss%=n/(N+n)×100`	SP=1 each; ratio directly	N=33,n=3 → 3/36 = **8⅓%**	**SP=1 per article, plug ratio**	Using n/N instead of n/(N+n)
Dishonest weight	False weight sold at CP	Gain %	"uses faulty/short weight"	`Gain% = Error/(True−Error) ×100`	Error over remaining true	Sells 960 g as 1 kg → 40/960 = **4⅙%**	**Error/(True−Error)**	Error/True (understates gain)

2 · Percentage 🔴 Feeds P&L, SI/CI, DI

Convert every % to a fraction (25%=1/4, 12½%=1/8, 16⅔%=1/6). The fraction method is 3× faster than long multiplication.

SITUATION	GIVEN	ASKED	RECOGNITION CUE	SHORTCUT / FORMULA	RATIO / FRACTION TRICK	WORKED EXAMPLE	FASTEST ROUTE	COMMON MISTAKE
Price ↑, cut use	Price rises x%	% cut in consumption	"expenditure unchanged"	`x/(100+x) ×100`	Rise 1/4 → cut 1/(4+1)=1/5	+25% → 25/125 = **20%**	**Fraction: rise 1/n → cut 1/(n+1)**	Reducing by the same 25%
A is x% more than B	A vs B relation	B is how much less	"by what % is B less than A"	`x/(100+x) ×100`	+25% ↔ −20% (5:4 pair)	A 25% more → B is **20%** less	**Memorize ± pairs (⅓↔¼, 25↔20)**	Assuming it is also 25%
Election margin	Winner w%, wins by V	Total votes	"won by … votes"	Margin= (2w−100)% ; `Total=V×100/margin`	V ÷ margin-fraction	60%, win 1600 → margin 20% → **8000**	**V/(margin%) ×100**	Dividing V by 60% not by margin
↑then↓ same x%	+x% then −x%	Net change	"increased then decreased"	`net = −x²/100 %` (always loss)	Factor 1.1×0.9 = 0.99	+10% −10% → **−1%**	**Multiply factors, or $-x^2/100$**	Concluding "no change"
Successive %	a% then b%	Single equivalent	two chained changes	`a + b + ab/100`	Multiply decimal factors	+20,+30 → 20+30+6 = **+56%**	**a+b+ab/100 (sign of b matters)**	Just adding a+b
Population back-track	+p%,−q% over years	Original	"present population is …"	P × factor_1 × factor_2 = now	Divide now by product of factors	×1.1×0.9=0.99; 29700/0.99 = **30000**	**Multiply factors, divide once**	Adding/subtracting % on final value

3 · Simple & Compound Interest 🔴 High Repeat

For 2 years, the whole chapter reduces to two memorized results. For rate-finding, read the amount ratio as a perfect square/cube.

SITUATION	GIVEN	ASKED	RECOGNITION CUE	SHORTCUT / FORMULA	RATIO TRICK	WORKED EXAMPLE	FASTEST ROUTE	COMMON MISTAKE
CI−SI, 2 yr	Difference, r, t=2	Principal	"difference between CI and SI, 2 years"	`Diff = P(r/100)²`	Diff = P × (r%)^2 directly	Diff 50, r 10% → P×0.01=50 → **5000**	**P = Diff×(100/r)^2**	Using 3-yr formula for 2 yr
CI−SI, 3 yr	Difference, r, t=3	Principal	"...for 3 years"	`Diff = P(r/100)²·(3+r/100)`	—	r 10%: Diff = P(0.01)(3.1)	**Plug once; watch the (3+r/100)**	Forgetting the (3+r/100) factor
Find rate (CI)	P → A in 2 yr	Rate	"amounts to ... in 2 years CI"	`A/P = (1+r/100)²`	Read A/P as a square	2420/2000 = 1.21 = (1.1)^2 → **10%**	**Recognize 1.21, 1.331, 1.44...**	Solving a full quadratic
Amount at two times	A at t and at 2t	Principal	"becomes X after 3 yr, Y after 6 yr"	Ratio Y/X = $(1+r)^t$; P=X/(that)	Y/X gives the growth factor for t	20070/13380 =1.5; P=13380/1.5 =**8920**	**Divide the two amounts first**	Trying to find r explicitly
Doubling / n-fold	Sum doubles in t yr (CI)	Time to 4×, 8×	"doubles ... becomes 4 times"	2× in t → 4× in 2t → 8× in 3t	Powers of the doubling time	2× in 5 yr → 8× in **15 yr**	**Multiply t by $\log_2$(fold)**	Linear scaling (4×→10 yr)

4 · Time & Work + Pipes & Cisterns 🔴 Every paper

LCM method beats fractions every time. Total Work = LCM(days). Efficiency = Total ÷ days → whole numbers → add/subtract freely.

SITUATION	GIVEN	ASKED	RECOGNITION CUE	SHORTCUT / FORMULA	LCM / RATIO TRICK	WORKED EXAMPLE	FASTEST ROUTE	COMMON MISTAKE
Two together	A=a days, B=b days	Combined time	"working together"	`ab/(a+b)`	LCM: add efficiencies, Total÷sum	12 & 18 → 216/30 = **7.2 d**	**Small numbers: ab/(a+b) mentally**	Averaging the two times
Find one alone	(A+B) & B	A alone	"together …, B alone …"	`1/A = 1/T − 1/B`	LCM: eff^{A} = Total − eff^{B}	T=8,B=12 → 1/8−1/12=1/24 → **24 d**	**Subtract efficiencies on LCM**	Subtracting the days directly
Pairwise (A+B,B+C,C+A)	3 pair-times	All-three time	"A&B in…, B&C in…, C&A in…"	Sum pairs = 2(A+B+C)	LCM: pair effs, halve the sum	12,15,20 → sum 1/5 → all = **10 d**	**Add three rates, halve**	Averaging the three pair-times
% more efficient	A is x% more than B	Other's time	"25% more efficient"	eff 125:100=5:4 → time 4:5	Time ratio = 100:(100+x)	A=20d (4 parts) → B = **25 d**	**Flip eff-ratio to time-ratio**	Adding 25% to A's days
Leave / join midway	Some work, then change	Remaining days	"after k days, X leaves/joins"	Units done → Total − done → ÷ new rate	Track units on the LCM board	B+C 3d=42u of 72 → A does 30/3=**10 d**	**Subtract units, divide by new eff**	Restarting the whole fraction sum
Men–Days–Hours	$M_1D_1H_1$, work scaled	New M/D/H	"×larger work, hrs/day"	$M_1D_1H_1/W_1 = M_2D_2H_2/W_2$	Keep man-days constant	10·15·8/1 = 12·D·10/3 → **30 d**	**One chain-rule equation**	Mixing men & days before final ÷
Pipe + leak	Fill x, empty y	Net fill time	"leak empties in …"	`Net = 1/x − 1/y` (inlet +, leak −)	LCM: +eff for fill, −eff for leak	10 & 15 → +3−2=+1 of 30 → **30 h**	**LCM board with signs**	Adding instead of subtracting leak

SITUATION	GIVEN	ASKED	RECOGNITION CUE	SHORTCUT / FORMULA	LCM / RATIO TRICK	WORKED EXAMPLE	FASTEST ROUTE	COMMON MISTAKE
Leak from delay	Fills in x, took x′ with leak	Leak-empty time	"due to leak took longer"	`Leak = x·x′/(x′−x)`	1/x − 1/x′ = leak rate	12 & 16 → 192/4 = **48 h**	**x·x′/(x′−x)**	Using x′−x as the answer

5 · Partnership

Profit share ∝ **capital × time**. Reduce to a clean ratio, then split. Deduct any salary/commission *before* ratio-splitting.

SITUATION	GIVEN	ASKED	RECOGNITION CUE	SHORTCUT / FORMULA	RATIO / BUTTERFLY	WORKED EXAMPLE	FASTEST ROUTE	COMMON MISTAKE
Equal time	Capitals, same period	Profit split	"invested for a year each"	Share ∝ capital	Ratio of capitals	C 3000:5000 = 3:5 of profit	**Cap ratio = profit ratio**	Forgetting to reduce ratio
Different times	Capital & months differ	Profit split	"A for 12 mo, B for 8 mo"	`Share ∝ C×T`	Multiply C×T, take ratio	C 4000×12 : 6000×8 = 48:48 = **1:1**	**Equivalent capital = C×T**	Using capital only, ignoring time
Work-done share (link)	Wages for a job	Each share	"earn ₹… for the work"	Wages ∝ work = eff×days	C does 4/11 → 4/11 of money	₹1540, C does 4/11 → **₹560**	**Work-fraction × total money**	Splitting by days agreed, not work
Working partner	Salary + capital ratio	Net split	"gets salary for managing"	Deduct salary → split remainder by C×T	Remaining ÷ equivalent capital	Profit−salary, then ratio	**Salary out first, ratio after**	Ratio-splitting the whole profit

6 · Mixture & Alligation

Alligation is the fastest tool whenever a **mean/average value** sits between two ingredient values — it gives the mix ratio in one cross.

SITUATION	GIVEN	ASKED	RECOGNITION CUE	SHORTCUT / FORMULA	ALLIGATION CROSS (BUTTERFLY)	WORKED EXAMPLE	FASTEST ROUTE	COMMON MISTAKE
Two-ingredient mix	Cheap c, dear d, mean m	Mix ratio	"mixed to cost/average …"	`cheap:dear = (d−m):(m−c)`	Cross-subtract m diagonally	Rice ₹20 & ₹30, mean 24 → (30−24):(24−20)=**6:4 =3:2**	**Draw the X, cross-subtract**	Reversing which diff goes with which
Milk–water dilution	Add water free	Ratio / profit	"adds water and sells at CP"	Water is 0-cost ingredient in cross	Treat water value = 0	Milk ₹40 to mean 32 → water:milk = (40−32):(32−0)=8:32 =**1:4**	**Alligation with 0 for water**	Forgetting water contributes 0 value
Repeated replacement	Remove & refill n times	Final pure qty	"drawn off and replaced, n times"	`Final = V(1−r/V)ⁿ`	Ratio (1−r/V) per operation	40L, 8L out ×3 → $40(4/5)^3$ = **20.48 L**	**Power formula once**	Subtracting r each time linearly
Two mixtures combined	Ratios in each vessel	Final ratio	"mix vessel A and B"	Find each part's fraction, alligate on the fraction	Cross on the milk-fraction	3:2 & 7:3 milk → alligate .6 & .7	**Alligation on the component fraction**	Adding the ratios directly

7 · Speed · Time · Distance 🔴 High Repeat

One conversion unlocks most sums: `km/h → m/s ×5/18` · `m/s → km/h ×18/5`.

SITUATION	GIVEN	ASKED	RECOGNITION CUE	SHORTCUT / FORMULA	RATIO TRICK	WORKED EXAMPLE	FASTEST ROUTE	COMMON MISTAKE
Equal-distance avg	Speeds x, y (same dist)	Average speed	"half journey at …, half at …"	`2xy/(x+y)` (harmonic)	Never the plain mean	40 & 60 → 4800/100 = **48**	**2xy/(x+y) straight**	Taking (x+y)/2 = 50 (trap)
Late / early	Two speeds, ±minutes	Distance	"late at …, early at …"	Time gap = (late+early); d/x − d/y = gap	Add late+early → total gap	5&6 kmh, 10 late+5 early=¼h → **7.5 km**	**Sum the times, one equation**	Subtracting late−early
Same-time meeting	Two bodies start together	Distances / total	"meet after starting together"	Distance ∝ speed (same time)	$d_1:d_2 = s_1:s_2$	20:30=2:3; 2 parts=60 → total **150 km**	**Speed ratio = distance ratio**	Assuming they cover equal distance
Ratio of speeds ↔ time	Fixed distance	Time ratio	"same distance, speeds a:b"	Time ∝ 1/speed	speeds a:b → times b:a	speeds 3:4 → times **4:3**	**Invert the speed ratio**	Keeping the same ratio for time

8 · Trains

Decide the **distance to cover** first: pole = train length only; platform/bridge = length + platform; passing another train = **sum** of both lengths.

SITUATION	GIVEN	ASKED	RECOGNITION CUE	SHORTCUT / FORMULA	RATIO TRICK	WORKED EXAMPLE	FASTEST ROUTE	COMMON MISTAKE
Cross a pole/man	L, t	Speed	"crosses a pole/standing man"	`S = L/t`	—	150 m in 15 s → 10 m/s = **36 km/h**	**L/t then ×18/5**	Adding a platform that isn't there

SITUATION	GIVEN	ASKED	RECOGNITION CUE	SHORTCUT / FORMULA	RATIO TRICK	WORKED EXAMPLE	FASTEST ROUTE	COMMON MISTAKE
Cross platform/bridge	L, P, t	Speed / length	"crosses a platform/bridge"	`S = (L+P)/t`	—	200+300 in 25 s → 20 m/s = **72 km/h**	**(L+P)/t**	Forgetting to add platform
Pole then platform	t_1 (pole), t_2 (platform P)	Length	"pole in …, platform in …"	Extra time (t_2-t_1) covers P → $S=P/(t_2-t_1)$	Platform ÷ extra time = speed	120 m in extra 6 s → 20 m/s, L=**240 m**	**$P/(t_2-t_1)$ gives speed instantly**	Setting up two full equations slowly
Two trains, opposite	L_1,L_2, speeds	Crossing time	"opposite directions"	Rel speed = S_1+S_2; dist = L_1+L_2	Add speeds, add lengths	120+80 m at 72 km/h(20 m/s) → **10 s**	**$(L_1+L_2)/(S_1+S_2)$**	Subtracting speeds
Two trains, same dir	L_1,L_2, speeds	Overtake time	"same direction / overtakes"	Rel speed = S_1-S_2; dist = L_1+L_2	Subtract speeds, add lengths	300 m at 12 km/h(10/3 m/s) → **90 s**	**$(L_1+L_2)/(S_1-S_2)$**	Using only one train's length

9 · Boats & Streams

Two back-solve identities crack almost everything: `b = (down+up)/2` and `s = (down−up)/2`.

SITUATION	GIVEN	ASKED	RECOGNITION CUE	SHORTCUT / FORMULA	RATIO TRICK	WORKED EXAMPLE	FASTEST ROUTE	COMMON MISTAKE
Basic up/down	b (still), s (stream)	Down/up speed	"downstream / upstream"	`down=b+s, up=b−s`	—	b 8, s 2 → down **10**, up 6	**Add for down, subtract for up**	Swapping the ± signs

SITUATION	GIVEN	ASKED	RECOGNITION CUE	SHORTCUT / FORMULA	RATIO TRICK	WORKED EXAMPLE	FASTEST ROUTE	COMMON MISTAKE
Back-solve b & s	down & up speeds/times	b or s	"…km down in …h, back in …h"	`b=(d+u)/2, s=(d−u)/2`	Convert each leg to speed first	15 & 9 → b=12, s=**3**	**Speeds first, then the two identities**	Averaging times, not speeds
Up = k × down time	Time ratio, s known	b	"twice/thrice as long upstream"	down speed = k × up speed → solve	(b+s)=k(b−s)	up=2×down time, s=3 → b=**9**	**Set (b+s)=k(b−s), solve b**	Equating times to speeds directly
There-and-back time	b, s, total time	Distance	"to a place and back in … h"	d/(b+s) + d/(b−s) = T	Sum the two leg-times	b6,s2,T2h → d/8+d/4=2 → **16/3 km**	**One equation, solve d**	Using one speed for both legs

10 · Ratio & Proportion

SITUATION	GIVEN	ASKED	RECOGNITION CUE	SHORTCUT / FORMULA	BUTTERFLY / TRICK	WORKED EXAMPLE	FASTEST ROUTE	COMMON MISTAKE
Combine A:B, B:C	Two chained ratios	A:B:C	"A:B=…, B:C=…"	Make B common (cross-multiply)	×B-terms across	2:3 & 4:5 → **8:12:15**	**Scale each to common B**	Just writing 2:3:5
Divide by ratio	Total, parts	One share	"divide ₹… in ratio …"	Each part = Total ÷ Σparts	Multiply share by part-value	7800 in 6:4:3(13) → C=3×600=**1800**	**One part value, then multiply**	Not clearing fraction ratios first

SITUATION	GIVEN	ASKED	RECOGNITION CUE	SHORTCUT / FORMULA	BUTTERFLY / TRICK	WORKED EXAMPLE	FASTEST ROUTE	COMMON MISTAKE
Proportionals	a,b (,c)	Mean/3rd/4th	"mean/third/fourth proportional"	mean √(ab); 3rd b^2/a; 4th bc/a	ad=bc cross-check	4,9 mean = √36 = **6**	**Pick the right proportional formula**	Confusing 3rd vs 4th proportional
Componendo-dividendo	a/b = c/d	(a+b)/(a−b)	fraction with ± on both	`(a+b)/(a−b)= (c+d)/(c−d)`	Add/sub numerator & denom	Speeds up complex ratio identities	**Apply C&D directly**	Expanding the whole fraction

11 · Average

SITUATION	GIVEN	ASKED	RECOGNITION CUE	SHORTCUT / FORMULA	RATIO / SHORTCUT	WORKED EXAMPLE	FASTEST ROUTE	COMMON MISTAKE
Add one value	avg of n, new value	New average	"one more is included"	Δavg = (new − old avg)/(n+1)	Shift, don't re-sum	avg27(5), add33 → +6/6=+1 → **28**	**Deviation ÷ new count**	Recomputing full sum every time
Consecutive numbers	First & last	Average	"consecutive / evenly spaced"	(first+last)/2	Middle term = average	12…16 → (12+16)/2 = **14**	**Just the middle term**	Summing all then dividing
Replace a member	n, old, new person	Age/weight change	"replaced by, average changes"	New−Old = n × Δavg	Total shift spread over n	avg ↑2 over 8 → newcomer +16	**n×Δavg = difference**	Ignoring the ×n multiplier

★ Master Recognition Table — "Which trick, how fast?"

Skim this the night before the exam. Left = the words in the question; right = the reflex tool.

IF THE QUESTION SAYS…	INSTANT TOOL	WHY IT'S FASTEST
"expenditure unchanged / consumption"	Fraction: rise 1/n → cut 1/(n+1)	No algebra, one fraction
"gain x%, find CP"	SP×100/(100+x) as a reduced fraction	Single multiply
"marks above & gives discount"	x − y − xy/100	Avoids MP→SP steps
"CI − SI for 2 years"	$P(r/100)^2$	Whole chapter → one line
"amounts to … in 2 yr CI"	Read A/P as a perfect square	No quadratic
"working together"	LCM board / $ab\cdot(a+b)^{-1}$	Whole-number efficiencies
"x% more efficient"	Flip eff-ratio → time-ratio	Ratio, not equations
"leak / pipe empties"	Signed LCM (+fill, −leak)	Add rates with sign
"invested for different months"	Equivalent capital = C×T	Butterfly the C×T
"mixed to average / mean value"	Alligation cross	One diagonal subtraction
"drawn off and replaced n times"	$V(1-r/V)^n$	Power beats step-by-step
"half at …, half at …" (speed)	2xy/(x+y)	Harmonic, dodges the mean trap
"late … early …"	gap = late+early, one equation	Straight to distance
"crosses pole vs platform"	Extra time × speed = platform	Speed in one subtraction
"downstream … upstream …"	b=(d+u)/2, s=(d−u)/2	Two-line back-solve
"start together & meet"	distance ∝ speed	Ratio, no time needed
"A:B and B:C"	Make B common (cross-multiply)	Butterfly across

Three universal speed-tools that reappear in every topic above:

◆ Butterfly / cross-multiplication — combining ratios, alligation, C×T partnership, proportion (ad=bc).

▣ **LCM board** — Time&Work, Pipes; turn every rate into a whole number.

⚖ **Ratio inversion** — efficiency↔time, speed↔time; whatever is "more" of one is "less" of the other in exact inverse.
